

FHRP: First Hop Redundancy Protocols

A Hands-On Guide to Redundant Routing

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0.1 Introduction to Network Topology for FHRP Implementation

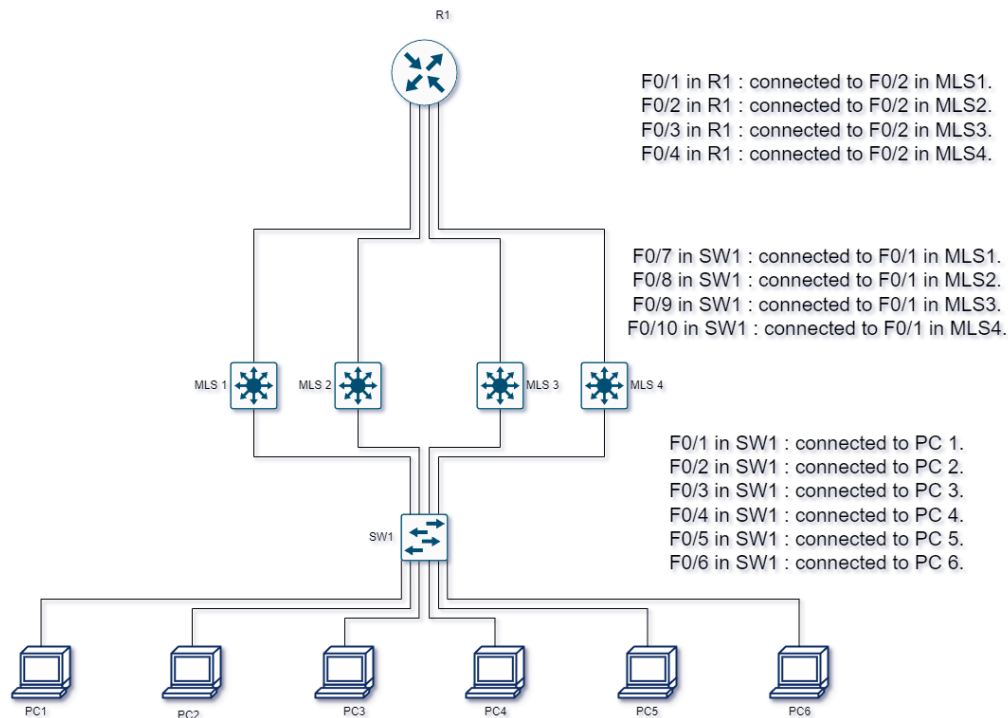


Figure 1:

This network topology is designed to demonstrate the implementation and functionality of First Hop Redundancy Protocols (FHRP), including **HSRP (Hot Standby Router Protocol)**, **VRRP (Virtual Router Redundancy Protocol)**, and **GLBP (Gateway Load Balancing Protocol)**. These protocols ensure high availability and fault tolerance by providing redundancy for default gateway routers in a network.

Topology Components

The topology consists of the following key components:

- **R1 (Router 1):** Serves as the upstream router that connects the network to external destinations.
- **MLS (Multi-Layer Switches) 1, 2, 3, and 4:** Act as Layer 3 distribution switches that participate in FHRP to provide redundancy and load balancing for end devices.
- **SW1 (Layer 2 Switch):** Functions as the access layer switch, connecting end devices (PC1 to PC6) to the network.
- **End Devices (PC1 - PC6):** Represent client machines that rely on a stable and resilient default gateway for communication beyond their local subnet.

Interconnections and Purpose

- The **end devices are connected to SW1** through FastEthernet ports (F0/1 to F0/6).
- SW1 is connected to the **multi-layer switches (MLS1 - MLS4)** via uplink ports (F0/7 to F0/10).
- Each **multi-layer switch is connected to the router (R1)** via FastEthernet links to provide redundancy at the Layer 3 distribution level.

Chapter 1

HSRP Hands-On Configuration

1.1 Introduction to HSRP Configuration

1.1.1 Overview of HSRP and Its Importance

Hot Standby Router Protocol (HSRP) is a Cisco proprietary redundancy protocol that provides **high availability** for IP networks by ensuring a backup router takes over automatically when the primary router fails.

In the **topology provided**, we have:

- R1 (Router)
- Four Multi-Layer Switches (MLS1, MLS2, MLS3, MLS4)
- One Layer 2 Switch (SW1)
- Six PCs (PC1 - PC6)

HSRP ensures **network uptime** by providing a **virtual gateway IP address**, which devices use as their default gateway. The **primary device** (Active Router) forwards packets, while the **secondary device** (Standby Router) takes over in case of failure.

1.2 Basic HSRP Setup

1.2.1 Configuring an Interface for HSRP

Each Multi-Layer Switch (MLS) needs an **HSRP virtual IP address** to act as the default gateway for connected hosts.

Lab Exercise 1: Configuring HSRP on VLAN 10

Objective: Configure **HSRP on VLAN 10** for MLS1 and MLS2 with a virtual IP.

Configuration Steps:

1. Access the VLAN interface on **MLS1** and assign an IP address.

[Click here to display the answer:](#)

2. Repeat for **MLS2** (standby switch).

3. [Click here to display the answer:](#)

4. Verify HSRP status.

[Click here to display the answer:](#)

Expected Outcome:

- One switch (MLS1 or MLS2) should be **Active**.
- The other switch should be **Standby**.
- The PCs should use 192.168.10.1 as the default gateway.

1.2.2 Assigning Virtual IP Addresses in HSRP

- The `standby 1 ip` command defines the **virtual IP address**.
- All hosts on **VLAN 10** should have 192.168.10.1 as their default gateway.

1.2.3 Verifying HSRP Configuration

To verify:

```
switch# show standby vlan 10
```

This command displays:

- **Active and Standby router roles**
- **Virtual IP and priority values**

1.3 HSRP Priority and Preemption

1.3.1 Setting HSRP Priority Levels

By default, the HSRP **Active Router** is chosen based on the highest **IP address**. However, we can manually assign priority.

Lab Exercise 2: Configuring HSRP Priority on MLS1 and MLS2

Objective: Set **MLS1** as the **Active Router** by increasing its priority.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- MLS1 should now be the **Active Router** because it has **higher priority (110 vs. 100)**.
- MLS2 should remain **Standby**.

1.3.2 Configuring Preemption for Faster Failover

- If the Active Router (**MLS1**) fails, **MLS2** takes over.
- When **MLS1** comes **back online**, it should **automatically reclaim the Active role**. This is done using **preemption**.

Lab Exercise 3: Configuring Preemption

Objective: Ensure **MLS1** **reclaims its role** as Active after recovery.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- If MLS1 fails, MLS2 becomes Active.
- When MLS1 returns, it **automatically reclaims** the role because of preemption.

1.3.3 Adjusting Preemption Delay Timers

To avoid flapping (switching back and forth between routers), we can add a **preemption delay** to wait before re-electing the Active Router.

Lab Exercise 4: Configuring Preemption Delay

Objective: Set a **30-second delay** before preemption occurs.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- **MLS1 will wait 30 seconds** before reclaiming Active status after recovering from failure.

1.4 HSRP Timer Adjustments

1.4.1 Modifying HSRP Hello and Hold Timers

By default, HSRP uses the following timers:

- **Hello Timer:** 3 seconds
- **Hold Timer:** 10 seconds

Adjusting these timers can **speed up failover** or **stabilize the network** in case of high traffic or latency.

Lab Exercise 5: Adjusting HSRP Timers

Objective: Configure **faster failover** by modifying **hello** and **hold timers** on VLAN 10.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- **Hello Timer = 1 second**
- **Hold Timer = 3 seconds**
- **Failover will happen faster** if the Active Router fails.

1.5 HSRP Authentication

HSRP authentication prevents **unauthorized devices** from participating in the election process, which enhances security.

1.5.1 Configuring Text-Based Authentication

Text authentication allows devices in the same group to verify each other using a **shared password**.

Lab Exercise 6: Configuring Text Authentication

Objective: Secure HSRP using a **shared password** (“SecureHSRP”).

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- Both switches will **only communicate** if they have the same authentication key.

1.5.2 Configuring MD5 Authentication

MD5 authentication encrypts the authentication key, making it more secure than plain text.

Lab Exercise 7: Configuring MD5 Authentication

Objective: Use **MD5 authentication** to secure HSRP communications.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- HSRP **will not function** if authentication keys don't match.

1.5.3 Configuring MD5 Key-Chain Authentication

Instead of storing a key as a plain string, we can use **Key-Chains**, which allow key rotation for added security.

Lab Exercise 8: Configuring Key-Chain Authentication

Objective: Use **MD5 key-chain authentication** for HSRP.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- HSRP uses **encrypted authentication** with rotating keys.

1.6 HSRP Tracking Mechanisms

HSRP tracking allows switches to **monitor interface states** and adjust priorities dynamically when an interface fails.

1.6.1 Tracking Interface Status with HSRP

When an uplink fails, the **standby router should take over immediately**.

Lab Exercise 9: Configuring HSRP Interface Tracking

Objective: Track interface **GigabitEthernet 1/1** on MLS1. If it fails, reduce priority by **20** to allow MLS2 to take over.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- If GigabitEthernet1/1 goes down, MLS1's priority decreases by **20**.
- If priority drops below MLS2, MLS2 will become **Active Router**.

1.6.2 Using Object Tracking for Failover

Instead of tracking interfaces, we can track an **IP route or other conditions**.

Lab Exercise 10: Configuring Object Tracking for HSRP

Objective: Track an **IP route (192.168.1.0/24)** and decrement HSRP priority if it becomes unreachable.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- If 192.168.1.0/24 route fails, MLS1 will lower its priority by **15**, allowing MLS2 to take over.

1.6.3 Adjusting HSRP Tracking Decrement Values

You can modify **how much priority is reduced** when tracking a failure. For example, a higher decrement value makes failover happen **faster**:

```
MLS1(config-if)# standby 1 track 1 decrement 30
```

Expected Outcome:

- If the tracked object **fails**, the priority is lowered more significantly, triggering a failover faster.

1.7 HSRP Versioning and IPv6

1.7.1 Upgrading to HSRP Version 2

By default, HSRP uses **Version 1**, but **Version 2** offers:

- Support for **larger group numbers** (0–4095 instead of 0–255)
- IPv6 compatibility
- Faster failover performance

Lab Exercise 11: Enabling HSRP Version 2

Objective: Upgrade HSRP to **Version 2** on MLS1 and MLS2.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- **HSRP version 2** should be active on both switches.
- **Virtual IP and group number remain unchanged.**

1.7.2 Configuring HSRP for IPv6

HSRP also works with **IPv6**, providing the same high-availability benefits.

Lab Exercise 12: Configuring HSRP for IPv6

Objective: Implement **IPv6 HSRP** on VLAN 20 using a **link-local virtual address**.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- IPv6 HSRP should be operational.
- **Virtual IPv6 address** should be auto-configured.

1.8 Verifying and Troubleshooting HSRP**1.8.1 Using show standby Commands for Verification**

The `show standby` command is essential for **monitoring HSRP status**.

Lab Exercise 13: Verifying HSRP Configuration

Objective: Check which router is active and standby in VLAN 10. **Verification Command:**

[Click here to display the answer:](#)

Expected Output Example:

```
Vlan10 Group 1 State: Active
Virtual IP: 192.168.10.1
Active Router: Local
Standby Router: 192.168.10.3
Priority: 110
Preemption: Enabled
```

- **Active Router:** MLS1 (local device).
- **Standby Router:** MLS2 (192.168.10.3).
- **Preemption enabled** to reclaim Active role when MLS1 recovers.

Other Useful Commands:**1. Show detailed HSRP information:**

```
MLS1# show standby vlan 10
```

2. Check timers and authentication settings:

```
MLS1# show standby vlan 10 detail
```

1.8.2 Common HSRP Misconfigurations and Fixes**Issue #1: Both Routers in Active State**

Problem: Both MLS1 and MLS2 are showing as **Active**. **Solution:**

- Ensure **HSRP authentication keys match**.

```
MLS1# show standby vlan 10 | include Auth
```

- If different, reconfigure authentication:

```
MLS1(config-if)# standby 1 authentication md5 key-string HSRP_MD5
MLS2(config-if)# standby 1 authentication md5 key-string HSRP_MD5
```

Issue #2: Standby Router Never Becomes Active

Problem: When MLS1 fails, **MLS2 does not take over**. **Solution:**

- Ensure **preemption is enabled**:

```
MLS2(config-if)# standby 1 preempt
```

- Check if interface tracking is **causing a low priority**.

```
MLS2# show standby vlan 10
```

If priority is **too low**, increase it:

```
MLS2(config-if)# standby 1 priority 120
```

Issue #3: Hosts Can't Reach the Virtual Gateway**Problem:** PCs cannot ping 192.168.10.1 (HSRP virtual IP). **Solution:**

- Check if VLAN interface is **up** on both switches:

```
MLS1# show ip interface brief | include Vlan10
```

- If it shows "**administratively down**", enable it:

```
MLS1(config)# interface vlan 10
MLS1(config-if)# no shutdown
```


Chapter 2

VRRP Hands-On Configuration

2.1 Introduction to VRRP

2.1.1 Overview of VRRP and Its Importance

The **Virtual Router Redundancy Protocol (VRRP)** is an open-standard redundancy protocol that provides high availability for **default gateways** by electing a **Master** and **Backup** router.

VRRP ensures that if the **Master Router** fails, a **Backup Router** takes over **seamlessly**, preventing network downtime.

In the **topology provided**, we have:

- Router (R1)
- Four Multi-Layer Switches (MLS1, MLS2, MLS3, MLS4)
- One Layer 2 Switch (SW1)
- Six PCs (PC1 - PC6)

VRRP provides **fault tolerance** by allowing multiple **Layer 3 switches** to participate in the election process while hosts continue using a single **virtual IP address** as the default gateway.

2.1.2 Basic VRRP Concepts and Terminology

- **Master Router**: The **active** VRRP router that handles traffic.
- **Backup Router**: The **standby** VRRP router that takes over if the Master fails.
- **Virtual Router**: The **group of devices** sharing a single **virtual IP**.
- **Virtual IP Address (VIP)**: The **gateway address** used by end devices.
- **VRRP Group**: Identifies multiple routers participating in the **same failover group**.

2.2 Basic VRRP Setup

2.2.1 Configuring an Interface for VRRP

Each **Multi-Layer Switch (MLS)** needs to participate in a **VRRP group** with a **virtual IP address** for failover.

Lab Exercise 1: Configuring VRRP on VLAN 10

Objective: Configure **VRRP on VLAN 10** for MLS1 and MLS2 with a **virtual IP**.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- One switch (MLS1 or MLS2) should become the **Master**.
- The other switch should be **Backup**.
- PCs should use **192.168.10.1** as their **default gateway**.

2.2.2 Assigning Virtual IP Addresses in VRRP

- The `vrrp 1 ip` command defines the **virtual IP address** shared by the VRRP group.
- Hosts in VLAN 10 will use this **virtual IP** (192.168.10.1) as their default gateway.

2.2.3 Verifying VRRP Configuration

To check VRRP status:

```
MLS1# show vrrp vlan 10
```

This command displays:

- **Master and Backup router roles**
- **Virtual IP and priority values**
- **VRRP timers**

2.3 VRRP Priority and Master Election

2.3.1 Understanding VRRP Priority Levels

VRRP uses a **priority system (1–255)** to determine the **Master Router**.

- **Default priority = 100**
- **Highest priority wins**
- **Master router** always has the highest priority

2.3.2 Configuring VRRP Priority for Master Selection

By default, the router with the **highest IP address** becomes **Master**, but you can manually set priority.

Lab Exercise 2: Configuring VRRP Priority on MLS1 and MLS2

Objective: Ensure **MLS1** becomes the **Master Router** by assigning it a higher priority.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- **MLS1** should now be the **Master Router** because its **priority is 110 (higher than MLS2's 100)**.
- **MLS2** should remain **Backup Router**.

2.3.3 VRRP Master Election Process

VRRP determines the Master using these steps:

1. **Highest priority wins.**
2. If equal, the **router with the highest IP address wins.**
3. If the Master fails, the **Backup takes over** automatically.

Verifying Master Router:

```
MLS1# show vrrp
```

Expected Output Example:

```
Vlan10 Group 1 State: Master  
Virtual IP: 192.168.10.1  
Priority: 110  
Preemption: Enabled  
Master Router: Local
```

2.4 VRRP Preemption and Timers

2.4.1 Enabling Preemption for Faster Failover

By default, **preemption is enabled** in VRRP. This means that if a **higher-priority router** comes back online, it will **automatically reclaim the Master role**.

Lab Exercise 3: Configuring VRRP Preemption

Objective: Ensure MLS1 automatically becomes Master when it recovers.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- If **MLS1 fails**, MLS2 will take over.
- When **MLS1 comes back online**, it will reclaim the Master role automatically.

2.4.2 Adjusting VRRP Advertisement Timers

VRRP uses these timers:

- **Advertisement Interval:** How often the Master sends VRRP hello messages (default 1 second).
- **Hold Time:** How long the Backup waits before assuming Master role (default 3 seconds).

Lab Exercise 4: Configuring Custom VRRP Timers

Objective: Reduce failover time by setting faster VRRP timers.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- Failover happens **within 1.5 seconds** instead of the default 3 seconds.

2.5 VRRP Authentication

2.5.1 Configuring Text-Based Authentication in VRRP

VRRP authentication ensures only **authorized routers** participate in the election.

Lab Exercise 5: Configuring VRRP Authentication

Objective: Secure VRRP using a shared password (“SecureVRRP”).

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- Only devices with the **correct authentication key** will communicate.

2.5.2 Implementing MD5 Authentication for VRRP

MD5 authentication is **more secure** because it encrypts the authentication key.

Lab Exercise 6: Configuring MD5 Authentication

Objective: Use **MD5 authentication** to secure VRRP communications.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- VRRP **will not function** if authentication keys don't match.

2.6 VRRP Object Tracking and Failover

2.6.1 Tracking Interface Status with VRRP

VRRP **can track interfaces** and **reduce priority** if a tracked interface fails. This ensures a **Backup Router** takes over immediately.

Lab Exercise 7: Configuring VRRP Interface Tracking

Objective: Track interface **GigabitEthernet 1/1** on MLS1. If it fails, reduce priority by **20**.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- If **GigabitEthernet1/1** goes down, MLS1's priority decreases by **20**.
- If priority drops below MLS2, MLS2 will become **Master Router**.

2.6.2 Using Object Tracking to Adjust Priority Dynamically

Instead of tracking interfaces, VRRP can track an **IP route** and adjust priority dynamically.

Lab Exercise 8: Configuring IP Route Tracking for VRRP

Objective: Track an **IP route (192.168.1.0/24)** and decrement VRRP priority if it becomes unreachable.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- If the **tracked route fails**, MLS1 will lower its priority by **15**, allowing MLS2 to take over.

2.6.3 Adjusting VRRP Tracking Decrement Values

You can modify **how much priority is reduced** when tracking a failure.

For example, a **higher decrement value** speeds up failover:

```
MLS1(config-if)# vrrp 1 track 1 decrement 30
```

Expected Outcome:

- If the tracked object **fails**, the priority is lowered more significantly, triggering a failover faster.

2.7 VRRP Versioning and IPv6 Support

2.7.1 Upgrading to VRRP Version 3

VRRP **Version 3** supports **IPv6** and provides better scalability.

Key Benefits of VRRPv3:

- **Supports IPv6**
- Allows **higher group numbers** (up to 4095)
- Improves **failover performance**

Lab Exercise 9: Upgrading VRRP to Version 3

Objective: Upgrade VRRP to **Version 3** on MLS1 and MLS2.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- VRRP **Version 3** should be active on both switches.
- Virtual IP and group number remain unchanged.

2.7.2 Configuring VRRP for IPv6 High Availability

VRRPv3 allows redundancy for **IPv6 networks**. Instead of a **virtual IPv4 address**, it assigns a **virtual IPv6 address**.

Lab Exercise 10: Configuring VRRP for IPv6

Objective: Implement **IPv6 VRRP** on VLAN 20 using a **link-local virtual address**.

Configuration Steps:

[Click here to display the answer:](#)

Expected Outcome:

- IPv6 VRRP should be operational.
- **Virtual IPv6 address (fe80::1)** should be reachable.

2.7.3 Verifying and Troubleshooting VRRPv3

To verify VRRP for IPv6:

```
MLS1# show vrrp all
```

Expected Output:

```
Vlan20 Group 1 State: Master
Virtual IPv6: fe80::1
Priority: 110
Master Router: Local
```

2.8 Verifying and Troubleshooting VRRP

2.8.1 Using show vrrp Commands for Verification

The show vrrp command is essential for **monitoring VRRP status**.

Lab Exercise 11: Verifying VRRP Configuration

Objective: Check which router is Master and Backup in VLAN 10.

Verification Command:

[Click here to display the answer:](#)

Expected Output Example:

```
Vlan10 Group 1 State: Master
Virtual IP: 192.168.10.1
Master Router: Local
Backup Router: 192.168.10.3
Priority: 110
Preemption: Enabled
```

- **Master Router:** MLS1 (local device).
- **Backup Router:** MLS2 (192.168.10.3).
- **Preemption enabled** to reclaim the Master role.

Other Useful Commands:**1. Show detailed VRRP information:**

```
MLS1# show vrrp vlan 10
```

2. Check timers and authentication settings:

```
MLS1# show vrrp vlan 10 detail
```

2.8.2 Common VRRP Misconfigurations and Fixes

Issue #1: Both Routers in Master State

Problem: Both MLS1 and MLS2 are showing as **Master**.

Solution:

- Ensure **VRRP authentication keys match**.

```
MLS1# show vrrp vlan 10 | include Auth
```

- If different, reconfigure authentication:

```
MLS1(config-if)# vrrp 1 authentication md5 key-string VRRP_MD5
MLS2(config-if)# vrrp 1 authentication md5 key-string VRRP_MD5
```

Issue #2: Backup Router Never Becomes Master

Problem: When MLS1 fails, MLS2 does not take over.

Solution:

- Ensure **preemption is enabled**:

```
MLS2(config-if)# vrrp 1 preempt
```

- Check if **interface tracking is causing a low priority**.

```
MLS2# show vrrp vlan 10
```

If priority is **too low**, increase it:

```
MLS2(config-if)# vrrp 1 priority 120
```

Issue #3: Hosts Can't Reach the Virtual Gateway

Problem: PCs cannot ping 192.168.10.1 (VRRP virtual IP).

Solution:

- Check if **VLAN interface is up** on both switches:

```
MLS1# show ip interface brief | include Vlan10
```

- If it shows "**administratively down**", enable it:

```
MLS1(config)# interface vlan 10
MLS1(config-if)# no shutdown
```